

David Reger

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EDUCATION

The Pennsylvania State University

August 2020 – May 2024

PhD in Nuclear Engineering

Thesis Project: *Large Eddy Simulation and Direct Numerical Simulation to Improve Near-Wall Modeling in Porous Media Models of Pebble Bed Reactors*

The Pennsylvania State University

August 2016 – May 2020

Bachelors of Science in Nuclear Engineering and Mechanical Engineering

Senior Capstone Project: *Determination of an initial core loading for the Westinghouse AP1000*

SKILLS

General: Computational Fluid Dynamics, Discrete Element Method, Structural Analysis, Computer-Aided Design, GPU Computing, High-Performance Computing, Reduced-order modeling

Languages: C/C++, Python, Matlab, Fortran, L^AT_EX

Simulation Tools: NekRS/Nek5000, MOOSE Framework, Pronghorn, Griffin, SAM, Project Chrono, Relap-7, Diablo, STAR-CCM+

Other Softwares: Paraview, VisIt, Creo Parametric, Solidworks, Fusion 360, Microsoft Office Tools

WORK EXPERIENCE

Gas-Cooled Reactor Methods Lead | Idaho National Laboratory

January 2025 – Present

- Manage 7 work packages totaling ~\$3,000,000 in funding within the Advanced Reactor Technologies Gas-Cooled Reactor Program focused on the modeling and simulation of gas-cooled reactors
- Collaborate with partners in industry, academia, and the regulator to determine appropriate work scope to address gas-cooled reactor research needs
- Represent the US gas-cooled reactor program in international validation and benchmarking collaborations

Multiphysics Researcher | Idaho National Laboratory

July 2024 – Present

- Validate numerical models of lower plenum mixing, bypass flow, and reactor cavity cooling system performance with experimental data from international collaborators
- Model accident scenarios for gas and salt-cooled pebble bed reactors in Pronghorn and SAM to provide reference models for the NRC
- Develop an overlapping domain coupling methodology within the MOOSE framework for coupling between Pronghorn porous media simulation and SAM system thermal hydraulics simulation

Fluid-Structure Interaction Intern | Lawrence Livermore National Laboratory

May 2023 – July 2023

- Developed a wrapper to integrate the Diablo structural analysis code into the Cardinal Multiphysics Framework
- Performed fluid-structure interaction simulations of a cylinder oscillating in flow
- Verified the accuracy of the coupled simulation with data from literature

Transient Multiphysics Analysis Intern | Idaho National Laboratory

May 2022 – August 2022

- Simulated pebble fuel cycling in a general pebble bed reactor using the discrete element method (DEM) in Project Chrono
- Analyzed the evolution of the bed packing throughout the fuel cycling process
- Determined pebble velocity profiles for use in monte-carlo burnup calculations

Transient Multiphysics Analysis Intern | Idaho National Laboratory

May 2021 – August 2021

- Developed a methodology to generate localized point kinetic parameters for high-temperature gas reactors (HTGRs) using Pronghorn and Griffin

- Created coupled thermal-hydraulics and neutronics models based on HTGRs currently in consideration
- Performed coupled load-following transient simulation to verify the performance of the use of localized point kinetic parameters

Thermal Analysis Intern | *Idaho National Laboratory*

May 2020 – August 2020

- Developed a simulation model for a reference pebble bed reactor using the Pronghorn porous-media code
- Modeled the primary coolant loop using Relap-7 system code and coupled to the Pronghorn core model
- Gained familiarity with Idaho National Lab's MOOSE framework

Test Engineering Intern | *Lockheed Martin Missiles and Fire Control*

May 2019 – August 2019

- Supported multiple programs through the design and integration of test fixtures and tools
- Performed optical test procedures for the F-35 electro-optical targeting system in collaboration with senior engineers
- Designed components to satisfy test specifications using Creo Parametric 4.0

JOURNAL PUBLICATIONS

- [1] A. Novak, R. Carlsen, S. Shunert, P. Balestra, D. Reger, R. Slaybaugh, and R. Martineau. "Pronghorn: A multidimensional coarse-mesh application for advanced reactor thermal hydraulics". In: *Nuclear Technology* 207 (July 2021), pp. 1015–1046.
- [2] R. Stewart, P. Balestra, D. Reger, and E. Merzari. "Generation of localized reactor point kinetics parameters using coupled neutronic and thermal fluid models for pebble-bed reactor transient analysis". In: *Annals of Nuclear Energy* 174 (Sept. 2022).
- [3] G. Guidicelli, A. Abou-Jaoude, A. Novak, P. Balestra, B. Feng, and D. Reger et. al. "The Virtual Test Bed (VTB) Repository: A Library of Reference Reactor Models Using NEAMS Tools". In: *Nuclear Science and Engineering* 197 (Jan. 2023).
- [4] T. Nguyen, D. Reger, D. Dai, E. Merzari, H. Yuan, B. Jackson, and G. Busco. "Large Eddy Simulation of Convective Heat Transfer in a Random Pebble Bed Using the Spectral Element Method". In: *ASME Journal of Heat and Mass Transfer* 145 (Dec. 2023).
- [5] D. Reger, E. Merzari, P. Balestra, S. Schunert, Y. Hassan, and S. King. "Direct Numerical Simulation and Large Eddy Simulation of a 67-Pebble Bed Experiment". In: *Nuclear Technology* (July 2023).
- [6] D. Reger, E. Merzari, P. Balestra, S. Schunert, Y. Hassan, and H. Yuan. "An improved pressure drop correlation for modeling localized effects in a pebble bed reactor". In: *Nuclear Engineering and Design* 403 (Mar. 2023).
- [7] D. Reger, E. Merzari, P. Balestra, S. Schunert, Y. Hassan, H. Yuan, Y. Lan, P. Fischer, and M. Min. "Pressure Drop Correlation Improvement for the Near-Wall Region of Pebble-Bed Reactors". In: *Nuclear Technology* 209 (Jan. 2023), pp. 90–104.
- [8] D. Reger, E. Merzari, P. Balestra, R. Stewart, and G. Strydom. "Discrete element simulation of Pebble Bed Reactors on graphics processing units". In: *Annals of Nuclear Energy* 190 (Sept. 2023).
- [9] D. Reger, E. Merzari, P. Balestra, and Y. Hassan. "Improving the modeling of near-wall interphase heat transfer in porous media models of Pebble Bed Reactors". In: *Nuclear Engineering and Design* 427 (Oct. 2024).
- [10] D. Reger, E. Merzari, T. Nguyen, Y. Lan, P. Fischer, and Y. Hassan. "A Study of the Transition to Turbulence in a Bed of 67 Spherical Pebbles". In: *Flow, Turbulence and Combustion* (Nov. 2024).
- [11] R. Stewart, P. Balestra, D. Reger, E. Merzari, and G. Strydom. "High-fidelity simulations of the run-in process for a pebble-bed reactor". In: *Annals of Nuclear Energy* 195 (Jan. 2024).
- [12] S. Okyay, D. Reger, V. Coppo Leite, E. Merzari, P. Balestra, and G. Strydom. "High-fidelity forced convection simulations of the University of Wisconsin–Madison air-cooled reactor cavity cooling system". In: *Nuclear Engineering and Design* 445 (2025).
- [13] H. Yuan, T. Nguyen, and D. Reger et. al. "High-Fidelity CFD Simulation of Mixed Convection and Forced Convection in a Pebble Bed Test Reactor Core". In: *Nuclear Technology* (2025).

CONFERENCE PUBLICATIONS

- [1] D. Reger, E. Merzari, P. Balestra, S. Schunert, and Y. Hassan. “Comparison of Pebble Bed Velocity Profiles Between High-Fidelity and Intermediate-Fidelity Codes”. In: *ICONE-28*. Aug. 2021.
- [2] D. Reger, E. Merzari, P. Balestra, and R. Stewart. “Improved Point Kinetic Parameters Generation for Pebble Bed Reactor Transient Simulation Using NEAMS Tools”. In: *American Nuclear Society Winter Meeting*. Dec. 2021.
- [3] P. Balestra, R. Stewart, D. Reger, S. Schunert, and J. Ortensi. “Gas Cooled Reactor Equilibrium Core Calculations Using NEAMS Tools.” In: *American Nuclear Society Winter Meeting*. Nov. 2022.
- [4] S. Okyay, E. Merzari, F. Kong, and D. Reger et. al. “HI-STORM Overpack and MPC-32 Thermal-Hydraulic Model with MOOSE Framework.” In: *American Nuclear Society Winter Meeting*. Nov. 2022.
- [5] D. Reger, E. Merzari, P. Balestra, R. Stewart, and G. Strydom. “Discrete element method investigation of reflector dimples for a general pebble bed reactor design.” In: *American Nuclear Society Winter Meeting*. Nov. 2022.
- [6] D. Reger, E. Merzari, H. Yuan, Y. Hassan, P. Balestra, and S. Schunert. “On the development of an improved drag correlation for the near-wall region of pebble beds.” In: *American Nuclear Society Summer Meeting*. June 2022.
- [7] D. Reger, E. Merzari, H. Yuan, Y. Hassan, S. King, K. Ngo, P. Balestra, and S. Schunert. “Large eddy simulation of a 67-pebble bed experiment”. In: *Advances in Thermal Hydraulics*. June 2022.
- [8] R. Stewart, D. Reger, P. Balestra, and G. Strydom. “Methodology for Determining the Run-In Scenario for a Pebble Bed Reactor.” In: *American Nuclear Society Winter Meeting*. Nov. 2022.
- [9] D. Dai, P. Vegendla, H. Yuan, L. Brockmeyer, A. Kraus, A. Tentner, and D. Reger. “Comparison of CFD Simulations and Porous Media Simulations of a Gas-Cooled Pebble Bed Core under Forced and Mixed Convection Conditions.” In: *NURETH-20*. Aug. 2023.
- [10] Y. Hassan, D. Reger, A. Kraus, and E. Merzari. “Challenge Problem 4: Building a Flexible Multiscale Framework for Core Modeling.” In: *NURETH-20*. Aug. 2023.
- [11] S. Okyay, E. Merzari, D. Reger, and V. Coppo Leite. “Validation and Comparison of HI-STORM Overpack and Thermal-Hydraulic Model with MOOSE and NekRS.” In: *NURETH-20*. Aug. 2023.
- [12] D. Reger, E. Merari, P. Balestra, Y. Hassan, and S. Lee. “High-Fidelity Simulation of Pebble Beds: Toward and Improved Understanding of the Wall Channeling Effect.” In: *NURETH-20*. Aug. 2023.
- [13] D. Reger and E. Merzari. “Direct numerical simulation of pebble beds: Analysis of the turbulent heat flux”. In: *Turbulence, Heat and Mass Transfer 10*. Sept. 2023.
- [14] H. Yuan, D. Shaver, T. Nguyen, E. Merzari, D. Reger, and et. al. “High-Fidelity CFD Simulation of Mixed Convection in a Pebble Bed Test Reactor Core.” In: *NURETH-20*. Aug. 2023.

HONORS AND AWARDS

Best Paper <i>NURETH-21 Conference</i>	2025
Mark Mills Award <i>American Nuclear Society</i>	2024
1st Place, FEDSM Flow Visualization Contest <i>FEDSM 2024 Conference</i>	2024
Best Paper <i>NURETH-20 Conference</i>	2023
Blasingame Memorial Graduate Fellowship in Engineering <i>Pennsylvania State University</i>	2023
Best Poster <i>Penn State Institute for Computational Data Science Symposium</i>	2022
Best Poster in Nuclear Research and Experiments <i>Idaho National Laboratory</i>	2022

Young Members Thermal Hydraulics Research Competition Winner <i>American Nuclear Society</i>	2022
Best Paper <i>ICONE-28 Conference</i>	2021
Graduate Scholarship for Excellence In Engineering Award <i>Pennsylvania State University</i>	2021
Best Poster in Nuclear Research and Experiments <i>Idaho National Laboratory</i>	2021
Nuclear Energy University Program Graduate Fellowship <i>DOE Office of Nuclear Energy</i>	2021
Monty Schulz Memorial Scholarship in Nuclear Engineering Award <i>Pennsylvania State University</i>	2020

PROFESSIONAL SOCIETIES

Member <i>American Nuclear Society</i>
Member <i>Alpha Nu Sigma Honor Society</i>
Member <i>Institute of Nuclear Materials Management, Penn State Chapter</i>